

The key idea behind the paper is to understand the impact of Android Go on mobile web performance. Google introduced Android Go for entry level smartphones, which had 200 million users by 2022. The authors perform an empirical analysis of Android Go's casual impact on user's web browsing experience. Their results show that under memory pressure, Android Go reduces the mean PLT, mean TTFB, and the page crash rate by up to 52.1%, 49.4%, and 91.8%, respectively, relative to its non-Go counterpart when loading Alexa top 100 Web pages. These effects are driven by performance improvements that occur when device is operating under memory pressure.

As Android Go runs fewer system processes, it tends to have more free memory and CPU cycles, resulting in these improvements. Using random forest, the authors show that in non GO, reducing memory and CPU overhead introduced by DOM shows most improvements in PLT, while for GO, reducing the number of iframes does the trick. The authors also show that deprioritizing delay tolerant system processes can reduce PLTs under memory pressure, and increasing the number of cores or their frequency can exhibit similar results.

The paper studies Alexa top websites, focusing on time to first byte and time to onload event firing as the main metrics, and introduces memory pressure using a custom app. The app allocates memory until a target pressure threshold is achieved (normal, low or high). All experiments are conducted after the required memory pressure has been established. The results show that Android Go led to 12.2x lower crash rate, with mean size of crashed pages being 1.3x greater than non crashed pages. Android Go also results in a lower mean TTFB, reducing it by 890ms. The estimated casual effect on mean page load time is 6.6 seconds when only successful pageloads under both variants are taken into account. To estimate the crashed pages, paper performs a regression analysis which shows a casual effect of 7.8 seconds, while a regression analysis with Heckman selection model shows a mean casual effect of 11.1 seconds.

Android Go has a consistently lower page crash rate under memory pressure compared to non Go. Memory pressure also negatively impacts TTFB, although Go results in a lower mean TTFB. Similar trends are seen with PLT, with larger absolute gap in Go vs non Go. The authors see no statistically significant difference between two different Android Go versions. Higher CPU frequency, however, consistently shows better performance. Lastly, the authors present opportunities for improvement, which include web page and browser level optimizations. Reducing the number of DOM elements for non Go and reducing the number of iframes for Go have the highest performance impact. Optimizations can also be made at the OS level, for instance by deprioritizing system processes and coordinated scheduling. Lastly hardware level optimizations can also help improve web performance.

The paper presents a thorough analysis of the impact of OS optimization on web performance and shows some interesting insights. It would be interesting to perform a similar comparison between an Android Go device and an older (albeit refurbished) flagship device in the same price bracket. It may be a better value for the consumers to opt for older high end devices than low end/Android Go devices. Such an initiative from the manufacturers may also encourage recycling

of older devices, which may otherwise end up in landfills.